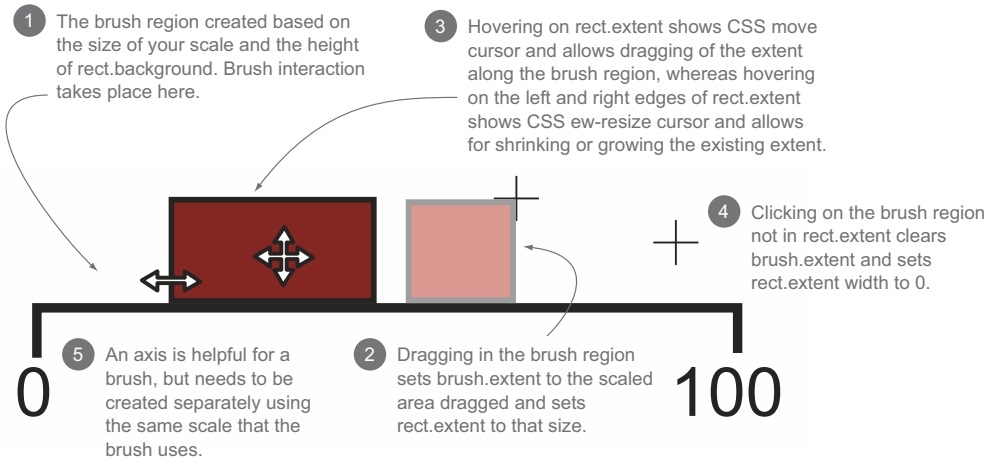


Figure 9.8 Components of a brush

event `"brush"` to call the function `brushed()`. Code to create the brush is shown in listing 9.14, while code for the actual behavior when the brush is used is explained in listing 9.15. The `"brush"` event happens any time the user drags the mouse along the brush region after clicking on the region.

Listing 9.14 Brush creation and function

```
function createBrush(incData) {
  var timeRange = d3.extent(incData.map(function(d) {
                return new Date(d.timestamp);
              })
  );
  var timeScale = d3.time.scale().domain(timeRange).range([10,990]);
  var timeBrush = d3.svg.brush()
                    .x(timeScale)
                    .extent(timeRange)
                    .on("brush", brushed);
  var timeAxis = d3.svg.axis()
    .scale(timeScale)
    .orient('bottom')
    .ticks(d3.time.hours, 2)
    .tickFormat(d3.time.format('%I%p'));
  var brushSVG = d3.select("#brush")
       .append("svg").attr("height", "100%").attr("width", "100%");
  brushSVG.append("g")
        .attr("transform", "translate(0,100)")
        .attr("id", "brushAxis").call(timeAxis);
  brushSVG.append("g").attr("transform", "translate(0,50)")
            .attr("id", "brushG").call(timeBrush)
            .selectAll("rect").attr("height", 50);
```

We'll make our brush dynamically resize later.

Gives us latest and earliest times of tweets

Fires the brushed() function on every "brush" event

Shows a tick for every two hours, and formats it to only show hour and AM/PM

Immediately calls the created rect elements and makes them 50 px high

Our axis

Our brush